

KARTIK SONI

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EDUCATION

B.S. Computer Science (Honors) · **University of Massachusetts Amherst (UMass Amherst)** · **GPA 3.9/4.0** Aug 2019 - May 2023

AWARDS & RECOGNITION

Cum Laude (Top 20%) · Dean's List (Multiple Semesters) · Bay State Scholarship · Employee of the Month & Team Awards @ Whova

TECHNICAL SKILLS

AI / ML	PyTorch · TensorFlow · Transformers · HuggingFace · LLMs · Fine-tuning · CNNs · NLP · Computer Vision OpenCV · Naïve Bayes · Cross-validation · Hyperparameter Tuning
Backend	Spring Boot · Java · Python · Django · Node.js · REST APIs · Redis · Microservices · JUnit · Mockito · Swagger
Frontend	Angular · React.js · TypeScript · JavaScript · HTML/CSS · MVVM · Lazy Loading · Reactive Forms
Practices	System Design · HLD/LLD · Agile/Scrum · Design Patterns · Code Review · PostgreSQL

EXPERIENCE

Software/ML Engineer · **Whova** *San Diego, CA, USA* | Aug 2023 - May 2026

- Owned POC coding, MVP scoping, Epic estimation, stakeholder communication, and cross-team delivery (PM, Design, QA, Infra) for an event management platform serving large-scale bookings.
- Worked on attendee engagement features such as Name Badges, Certificates, Post Event Report, and Call For Speakers.
- Built and shipped premium/enterprise add-ons supporting \$500k in annual revenue, doubled feature adoption (23% to 46%).
- Developed web apps using Angular, Django, PostgreSQL, Redis, RabbitMQ, Celery, S3, Gitlab, Jenkins, Datadog, Sentry.
- Reduced support tickets by 7% by building a RAG chatbot that answers organizers setup questions using LangChain, pgvector, hybrid retrieval and reranking.
- Built an AI agent to simplify multi-step setup process for call for speakers with a conversational flow (LangGraph, Pydantic), validating config constraints.
- Benchmarked open-source encoder-decoder models on 30 held-out reports scored by 5 reviewers on a custom rubric, then fine tuned the winner with LoRA (Hugging Face PEFT) on 10k curated event reports.
- Deployed self-hosted via vLLM on KServe(GKE) for data privacy, with Langfuse tracing and output monitoring – driving a ~15% increase in organizer engagement.
- Recognised as Employee of the Month for engineering innovation; mentored 3 interns/new engineers through onboarding, Agile practices, code reviews, and project bootstrapping.

ML Researcher · **UMass Amherst - Prof. Jaime Davila** *Amherst, MA, USA* | Feb 2023 - May 2023

- Designed and trained CNN architectures for medical image classification (disease detection) on datasets of 4 000 - 13 000 images, iterating on kernel sizes, depth, and regularisation to combat overfitting.
- Applied K-fold cross-validation and systematic hyperparameter search (learning rate, depth, optimiser), switching to SGD and achieving a final F1 score of 91%.
- Improved dataset quality by sourcing diverse, well-distributed images and applying augmentation (rotations, flips, brightness/contrast shifts), boosting generalisation on held-out sets.

Teaching Assistant - Systems Programming · [UMass Amherst](#) Amherst, MA, USA | Sep 2022 - May 2023

- Built a transformer-based grading system using BERT + sentence-transformers to evaluate semantic similarity, extract key concepts (NER), and generate feedback for a Systems Programming course. Achieved 92% agreement with human graders across 280+ assignments, reducing grading time from 4 hours/week to 20 minutes/week.
- Supported 100+ students on concurrency, synchronisation, pointers, dynamic memory, and control flow; designed assignments and labs that consistently received positive feedback.

Software Engineer Intern · [ISO New England](#) Holyoke, MA, USA | Jun 2022 - Aug 2022

- Migrated a legacy C++ application to a Spring Boot Java microservice; added JWT-based auth, role-based access, and Redis caching, reducing average API latency noticeably.
- Integrated SLF4J + Sentry for structured error logging; applied Lombok and SonarLint to raise code quality; scheduled automated emails via SendGrid cron jobs.

HOBBY PROJECTS

Neural Network Training Observability Tool · NumPy · PyTorch

- Built a per-layer training observability tool in PyTorch using forward hooks that detects exploding/vanishing gradients/activations and dead neurons from layer statistics (activation mean/std, dead fraction, gradient norms) and recommends targeted fixes - LeakyReLU, weight re-initialization, learning rate reduction - based on each failure's layer-wise signature.
- Closed the loop on MNIST digit classification and sentiment analysis (learned embeddings) by deliberately inducing each pathology, confirming correct diagnosis, applying the recommended fix, and verifying training recovery.
- Developed a from-scratch NumPy reference implementation of the complete training stack - forward pass + backpropagation via chain rule, gradient descent, linear regression through multi-layer perceptron, MSE and categorical cross-entropy losses, sigmoid/ReLU/softmax activations - verified with finite-difference gradient checks.
- Implemented the modern training stabilizer toolkit: Xavier/Kaiming weight initialization, Dropout, and the full normalization family (BatchNorm with running statistics and train/eval modes, LayerNorm, RMSNorm), demonstrating the statistical invariants each enforces.

Smart Home IoT System · Raspberry Pi · Python · Node.js · Angular · PostgreSQL

- Built a Raspberry Pi temperature/humidity monitor (DHT11) with sensor data streamed to Supabase PostgreSQL via a Node.js API; deployed an Angular dashboard on Vercel for real-time room visualisation and remote relay control.
- Added Pi-hole DNS-level ad blocking and relay-based scheduled lighting control, demonstrating edge AI/IoT system design end-to-end.

Real-Time Mask Detection (CNN) · TensorFlow · Keras · OpenCV · Python

- Trained a CNN on a manually labelled, augmented dataset (rotations, flips, brightness/contrast shifts, zoom) to achieve 92% real-time mask-detection accuracy.
- Deployed as an open-source initiative during COVID using OpenCV for live video inference; contributed to public safety tooling adopted by multiple organisations.

RELEVANT COURSE WORK

AI – Machine Learning, Computer Vision, Data Structures & Algorithms, NLP, Search Engine, Operating Systems, Computer Network, Programming Methodology, Reasoning under uncertainty